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A reduced viscoelastic FDTD formulation for ultrasound-driven shear wave propagation in soft tissue

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Abstract

Ultrasound-driven shear wave propagation in soft tissue underlies shear wave elastography (SWE) and emerging elastomechanical hypotheses of ultrasonic neuromodulation. Accurate modeling of radiation-force–induced tissue motion is therefore of central importance. However, general-purpose finite element elastodynamic solvers are often computationally expensive and unnecessarily broad for shear-dominant ultrasound applications. We derive a reduced viscoelastic formulation by systematically applying near-incompressibility, small-strain linearization, Helmholtz decomposition, and solenoidal force projection to the full Navier equations, yielding a Kelvin–Voigt shear wave equation that retains only the transverse dynamics relevant to radiation-force–induced motion. An explicit finite-difference time-domain (FDTD) implementation is developed with second-order accuracy in space and time. The solenoidal constraint on the body force is enforced via a matrix-free conjugate gradient (CG) Poisson solve; for the practically important case of separable radiation-force sources, a pre-computed projection reduces this cost by one to two orders of magnitude. Validation studies in homogeneous media demonstrate recovery of the theoretical shear wavespeed to within $<0.1\%$ (linear-fit estimate on a 180^3 grid), second-order convergence under grid refinement (clean trace- L_2 self-convergence, $p \gtrsim 2$), and agreement with analytical Kelvin–Voigt attenuation and phase speed predictions to within $\sim 3\%$ and $<0.5\%$, respectively, over a 16-point viscosity sweep ($\eta \in [0, 1.5]$ Pa·s). An end-to-end benchmark against the RSNA QIBA elastic phantom standard recovers shear wave speeds to within $\sim 1\%$ across a tenfold range of shear moduli ($G = 1\text{--}10$ kPa). The reduced FDTD framework provides a lightweight, physically transparent alternative to full multiphysics elasticity solvers that accommodates spatial heterogeneity in shear modulus, density, and viscosity and facilitates integration with acoustic simulation tools. End-to-end transcranial demonstrations through realistic micro-CT skull geometries are presented using two transducer configurations, producing shear displacements of $1\text{--}3\text{ }\mu\text{m}$ consistent with clinical ARFI measurements.

1 Introduction

Ultrasound-driven mechanical phenomena in soft tissue play a central role in several biomedical applications, most prominently shear wave elastography (SWE) [1, 2, 3, 4] and ultrasonic neuromodulation [5, 6, 7]. In SWE, acoustic radiation force [8, 9, 10, 11] generates transient elastic displacements whose propagation encodes the underlying shear modulus of tissue. In ultrasonic neuromodulation, related elastomechanical interactions have been proposed as potential mechanisms linking acoustic exposure to neural activity [5, 12, 13]. These application domains share a common requirement: accurate and computationally efficient modeling of elastic wave generation and propagation in soft solids under ultrasound excitation.

Extensive prior work has developed high-fidelity simulation frameworks for SWE based on general-purpose finite element methods (FEM) [14, 15, 16], including widely used packages such as LS-DYNA and Abaqus [17]. These tools can represent complex geometries, heterogeneous materials, nonlinear constitutive behavior, and boundary interactions, making them highly versatile. However, this generality comes at substantial computational and algorithmic cost. For ultrasound-specific

problems the primary interest lies in small, radiation-force-induced shear displacements in nearly incompressible soft tissue where the full generality of large-scale FEM is often unnecessary and can obscure the essential physics.

A central challenge in ultrasound-driven elastodynamics is the strong separation of physical scales between acoustic and shear phenomena. Acoustic waves in soft tissue propagate with speeds on the order of 10^3 m/s [18, 19], governed by the large bulk modulus, whereas shear waves travel at speeds typically near 1–5 m/s [20, 21]. This disparity produces temporal scales that differ by approximately three orders of magnitude. As a result, practical modeling strategies commonly treat the acoustic field and the resulting radiation-force-driven elastic response on separate timescales, coupling them through intermediate source terms: the acoustic pressure or intensity is computed first, the radiation force is derived from it, and that force is then supplied as a body force to a shear wave solver [22]. Such sequential acousto-elastic coupling approaches are well documented in the SWE literature [17, 2]. Full FEM modeling of ARF-induced shear wave propagation is covered by established guidelines [16], which describe a sequential pipeline and recommend commercial codes. That approach is general but computationally heavy.

Numerical simulation is widely used to validate and improve shear wave imaging techniques. FEM models have been developed for ARFI imaging [23], for the shear wave response to radiation force in breast [24] and liver [25], for viscoelastic propagation [26], and for thermal effects in soft tissue [27]. Alternatives to full-field FEM include the finite difference method [28, 29], the pseudospectral method [30], and the Green's function approach [31, 17, 32, 33]. Green's function methods are well suited when the response is needed at only a few observation points; full-field methods (FEM, FDTD, pseudospectral) instead compute the entire transient response over the domain.

Among finite-difference approaches, dedicated 3D FDTD simulations of shear waves for complex modulus imaging have been implemented [28]. That work assumes incompressibility and pure shear, yielding six first-order velocity-stress equations in 3D with Kelvin-Voigt viscoelasticity on a staggered grid with PML boundaries; the source is a vibrating needle chosen to minimize compressional energy. The present formulation, by contrast, uses a displacement-based update with an explicit solenoidal projection of the body force $\mathbf{f}_S$ via a scalar Poisson (CG) solve, so that arbitrary (e.g., radiation-force) sources can be applied without prescribing a needle geometry. Both are reduced shear-only forward solvers: the Orescanin et al. approach targets algebraic Helmholtz inversion for complex modulus reconstruction from harmonic excitation, whereas the present solver emphasizes integration with acoustic simulators and Kelvin-Voigt attenuation in the shear equation.

Green's function methods offer a different trade-off. GPU-accelerated implementations [22] provide fast evaluation of shear waves from a prescribed radiation force in elastic and viscoelastic media. The present FDTD formulation instead advances the full displacement field on a grid at each time step, so that the response is available everywhere at once and the same framework can accommodate spatial heterogeneity and general boundary conditions.

The objective of the present work is thus to derive and implement a reduced, purpose-built formulation of shear elastodynamics tailored specifically to ultrasound-driven problems in soft solids. Rather than pursuing full general-purpose elasticity, we seek to isolate the minimal physical model required to accurately reproduce radiation-force-induced shear displacements and their propagation, while reducing computational complexity. The formulation accommodates spatial heterogeneity in shear modulus, density, and viscosity. Validation is performed against analytical benchmarks for wavespeed recovery, grid convergence, and Kelvin-Voigt attenuation, as well as against the standardized RSNA QIBA elastic phantom [16]. This reduction enables tight integration with acoustic simulation tools and supports efficient multiphysics workflows relevant to SWE, elastomechanical hypotheses of ultrasonic neuromodulation, and iterative methods for inverse problems.

2 Reduced Elastodynamic Formulation

This section derives the governing equations for ultrasound-driven shear dynamics in soft tissue, beginning from the full linear elastodynamic continuum description and systematically applying the physical reductions described below. The main assumptions are near-incompressibility of soft tissue ($K \gg \mu$), small-strain linearization, and timescale separation between acoustic and shear dynamics.

2.1 Near-incompressibility in soft tissue and modulus hierarchy

A key simplification exploited in ultrasound-driven elastodynamics is the extreme separation between volumetric (compressional) stiffness and shear stiffness in soft tissue [34, 19]. In isotropic linear

elasticity [35, 36, 37], the longitudinal and shear wave speeds satisfy

$$c_L^2 = \frac{K + \frac{4}{3}\mu}{\rho}, \quad (1)$$

$$c_S^2 = \frac{\mu}{\rho}, \quad (2)$$

where ρ is density, K is bulk modulus, and μ is shear modulus. Using typical ultrasound values in soft tissue [19] ($c_L \approx 1540$ m/s and $\rho \approx 1000$ kg/m³) gives

$$K \approx \rho c_L^2 \approx 2.4 \times 10^9 \text{ Pa}, \quad (3)$$

where the correction $\frac{4}{3}\mu$ is negligible because $\mu \ll K$. Meanwhile, typical shear-wave speeds in elastography [20, 2] ($c_S \approx 1\text{--}5$ m/s) imply

$$\mu = \rho c_S^2 \approx 1 \times 10^3\text{--}2.5 \times 10^4 \text{ Pa}. \quad (4)$$

Thus, in ultrasound-relevant soft tissue,

$$\frac{K}{\mu} \sim 10^5\text{--}10^6, \quad (5)$$

which is a far stronger hierarchy than one would infer from moderate values of Poisson's ratio alone.

For completeness, the elastic constants satisfy

$$E = 2\mu(1 + \nu), \quad \nu = \frac{3K - 2\mu}{2(3K + \mu)}, \quad K = \frac{E}{3(1 - 2\nu)}. \quad (6)$$

Under the hierarchy $K \gg \mu$, these reduce to the familiar incompressible limits

$$\nu \approx \frac{1}{2} - \frac{\mu}{6K}, \quad E \approx 3\mu, \quad (7)$$

so that the Young's modulus remains in the kPa range even though the bulk modulus is in the GPa range.

Interpretation for ultrasound-driven shear wave propagation. The hierarchy $K \gg \mu$ implies that compressional disturbances propagate rapidly and are energetically expensive in volumetric strain, whereas the radiation-force-induced response of interest in elastography [20, 21] is dominated by slower shear motion governed primarily by μ . This motivates (i) shear-focused elastic models and (ii) sequential acousto-elastic workflows where the acoustic field is used to compute forcing, and the subsequent shear dynamics are simulated on a much slower time axis. In ultrasound-driven problems, the external body force arises from acoustic radiation force. Because the acoustic field evolves on timescales three orders of magnitude faster than shear propagation, the force distribution may be determined independently, via measurement, calculation, or computed using an acoustic simulation and treated as a prescribed source term in Eq. (16).

2.2 Full linear viscoelastodynamics

Let $\mathbf{u}(\mathbf{x}, t)$ denote the displacement field in an isotropic, linearly viscoelastic continuum of density ρ . We adopt a Kelvin–Voigt constitutive law [38, 39], in which the Cauchy stress is decomposed into an elastic and a viscous contribution,

$$\boldsymbol{\sigma} = \lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2\mu\boldsymbol{\varepsilon} + 2\eta\dot{\boldsymbol{\varepsilon}}, \quad (8)$$

where $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$ is the small-strain tensor, λ and μ are the Lamé parameters, and η is the shear viscosity. Conservation of linear momentum with body force density $\mathbf{f}$ yields

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}. \quad (9)$$

Substituting Eq. (8) into Eq. (9) gives

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu\nabla^2 \mathbf{u} + \eta\nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}, \quad (10)$$

where the viscous term contributes a Laplacian of particle velocity.

2.3 Helmholtz decomposition of displacement

The displacement field may be decomposed into irrotational (longitudinal) and solenoidal (shear) components [35, 36],

$$\mathbf{u} = \nabla\phi + \nabla \times \mathbf{\Psi}, \quad (11)$$

where ϕ is a scalar potential and $\mathbf{\Psi}$ is a vector potential satisfying $\nabla \cdot \mathbf{\Psi} = 0$.

Substituting into Eq. (10) yields decoupled wave equations for the longitudinal and shear components:

$$\frac{\partial^2 \phi}{\partial t^2} = c_L^2 \nabla^2 \phi + \frac{1}{\rho} f_L, \quad (12)$$

$$\frac{\partial^2 \mathbf{\Psi}}{\partial t^2} = c_S^2 \nabla^2 \mathbf{\Psi} + \frac{1}{\rho} \mathbf{f}_S, \quad (13)$$

where f_L and $\mathbf{f}_S$ denote the longitudinal and solenoidal projections of the body force.

2.4 Incompressible limit and reduced shear equation with viscosity

Given the extreme modulus hierarchy $K \gg \mu$ (Sec. 2.1), the incompressible limit is taken:

$$\nabla \cdot \mathbf{u} = 0, \quad (14)$$

which eliminates longitudinal dynamics and introduces a pressure field p as a Lagrange multiplier that enforces the constraint. In this limit, Eq. (10) becomes

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = -\nabla p + \mu \nabla^2 \mathbf{u} + \eta \nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}. \quad (15)$$

Taking the divergence of this momentum equation and enforcing Eq. (14) yields a Poisson equation for p that keeps the displacement divergence-free at each time. Projecting Eq. (15) onto the divergence-free subspace removes the pressure gradient and gives the reduced forced shear wave equation

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \mu \nabla^2 \mathbf{u} + \eta \nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}_S, \quad (16)$$

where $\mathbf{f}_S$ is the solenoidal projection of the applied body force. Equation (16) governs shear wave propagation with wavespeed $c_S = \sqrt{\mu/\rho}$ and frequency-dependent attenuation set by η . Under the assumptions of linear elasticity, near incompressibility, small displacement, and timescale separation between acoustic and shear dynamics, this equation forms the physical basis for the numerical formulation of ultrasound-driven tissue motion.

3 Numerical Methods

3.1 Explicit FDTD update

Let $\mathbf{u}^n$ denote displacement at time $t_n = n\Delta t$. The implemented update [40] is

$$\mathbf{u}^{n+1} = 2\mathbf{u}^n - \mathbf{u}^{n-1} + \Delta t^2 \mathbf{a}^n, \quad (17)$$

with acceleration

$$\mathbf{a}^n = \frac{\mu}{\rho} \nabla_h^2 \mathbf{u}^n + \frac{\eta}{\rho} \nabla_h^2 \dot{\mathbf{u}}^n + \frac{1}{\rho} \mathbf{f}_S^n, \quad \dot{\mathbf{u}}^n \approx \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t}. \quad (18)$$

∇_h^2 is a second-order finite-difference 3D Laplacian (7-point stencil) on cell centers.

For homogeneous media without viscosity, a standard stability bound is

$$\Delta t \lesssim \frac{1}{c_s} \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2} \right)^{-1/2}, \quad (19)$$

which motivates the CFL-based [41] step choice in validation scripts. The overall scheme is second-order accurate in time for the elastic component and first-order accurate in time for viscous dissipation, consistent with Kelvin–Voigt damping [38]. Numerical convergence and stability are verified empirically in Sec. 4 through grid-refinement and timestep studies.

3.2 Helmholtz projection via conjugate gradient

The reduced shear equation (16) requires the solenoidal projection $\mathbf{f}_S$ of the applied body force. In the general solver, a different body force can be supplied at each time step, and the solenoidal projection is computed independently for each one. This captures fully time-dependent body forces, such as those arising from a moving acoustic focus or time-varying pulses.

3.2.1 Solenoidal projection of body force. At each time step t_n , the body force $\mathbf{f}^n$ is decomposed into solenoidal and irrotational parts:

$$\mathbf{f}^n = \mathbf{f}_S^n + \nabla \phi^n, \quad \nabla \cdot \mathbf{f}_S^n = 0. \quad (20)$$

Taking divergence gives a scalar Poisson equation

$$\nabla^2 \phi^n = \nabla \cdot \mathbf{f}^n, \quad (21)$$

from which the solenoidal component is recovered as

$$\mathbf{f}_S^n = \mathbf{f}^n - \nabla \phi^n. \quad (22)$$

This projection removes forcing components that would excite compressional motion. In other words, the solver is a shear-mode propagator, not a full elastic-wave solver.

3.2.2 Choice of conjugate gradient for the scalar Poisson problem. Only one scalar field ϕ must be solved at each step (rather than a full vector system). After finite-difference discretization, this becomes a large sparse linear system $A\phi = b$, where A is the discrete negative Laplacian on interior nodes. Conjugate gradient (CG) [42, 43] is a natural choice because A is symmetric positive-definite on the interior unknowns, matching the theoretical requirements of CG. The 3D problem is large and sparse, making direct factorization memory-prohibitive and CG requires only repeated stencil-like matrix–vector operations, which scale well in three dimensions.

3.2.3 Numerical implementation details. The projection solve is performed in matrix-free form as

$$A\phi = b, \quad A = -\nabla_h^2, \quad b = \nabla_h \cdot \mathbf{f}, \quad (23)$$

where ∇_h^2 is a second-order central-difference 3D Laplacian (7-point stencil on interior nodes).

The code uses an interior mask m (1 in interior, 0 on outer layer) and applies it after each update:

$$\phi \leftarrow \phi \odot m, \quad r \leftarrow r \odot m, \quad p \leftarrow p \odot m. \quad (24)$$

This enforces the chosen boundary treatment and avoids boundary unknown drift.

Starting from $\phi_0 = 0$:

$$r_0 = (b - A\phi_0) \odot m, \quad p_0 = r_0, \quad (25)$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \quad (26)$$

$$\phi_{k+1} = (\phi_k + \alpha_k p_k) \odot m, \quad (27)$$

$$r_{k+1} = (r_k - \alpha_k A p_k) \odot m, \quad (28)$$

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}, \quad (29)$$

$$p_{k+1} = (r_{k+1} + \beta_k p_k) \odot m. \quad (30)$$

No sparse matrix is assembled explicitly; Aq is computed by applying the Laplacian stencil to q .

The solver terminates when relative residual drops below tolerance,

$$\sqrt{\frac{\|r_k\|_2^2}{\|r_0\|_2^2}} < \text{tol}, \quad (31)$$

or when maximum iterations are reached. Additional guards stop iterations if denominator or residual becomes non-finite or non-positive.

3.3 Pre-computed projection for separable body forces

In the common case of acoustic radiation force (ARF) driven shear wave excitation, the spatial and temporal dependences of the body force are separable. The ARF spatial distribution $\mathbf{b}_0(\mathbf{x})$ is determined entirely by the acoustic intensity field and tissue absorption, while the temporal dependence is controlled by the push envelope. The body force then takes the form

$$\mathbf{f}(\mathbf{x}, t) = \mathbf{b}_0(\mathbf{x}) g(t), \quad (32)$$

where $g(t)$ is a scalar temporal envelope (for example, a step function that equals one during the push and zero afterwards). Because the Helmholtz–Hodge decomposition is linear, the solenoidal projection commutes with the scalar envelope:

$$\mathbf{f}_S^n = \mathcal{P}_S[\mathbf{b}_0 g(t_n)] = g(t_n) \mathcal{P}_S[\mathbf{b}_0] = g(t_n) \mathbf{f}_{S,0}, \quad (33)$$

where $\mathbf{f}_{S,0} \equiv \mathcal{P}_S[\mathbf{b}_0]$ is the divergence-free projection of the spatial pattern.

3.3.1 Algorithm. The pre-computed projection proceeds in two stages:

1. **Pre-computation.** For a given spatial body force pattern $\mathbf{b}_0$, solve the scalar Poisson problem

$$\nabla_h^2 \phi_0 = \nabla_h \cdot \mathbf{b}_0, \quad (34)$$

via conjugate gradient, then form the solenoidal projection

$$\mathbf{f}_{S,0} = \mathbf{b}_0 - \nabla_h \phi_0. \quad (35)$$

This step is repeated each time the spatial force distribution changes.

2. **Time stepping.** While the spatial pattern remains fixed, the solenoidal force at each step n is obtained by scalar multiplication:

$$\mathbf{f}_S^n = g(t_n) \mathbf{f}_{S,0}. \quad (36)$$

The FDTD update then proceeds with the acceleration

$$\mathbf{a}^n = \frac{\mu}{\rho} \nabla_h^2 \mathbf{u}^n + \frac{\eta}{\rho} \nabla_h^2 \dot{\mathbf{u}}^n + \frac{g(t_n)}{\rho} \mathbf{f}_{S,0}. \quad (37)$$

In the standard ARF scenario the acoustic intensity field is computed once and the resulting spatial force pattern $\mathbf{b}_0(\mathbf{x})$ is fixed for the entire simulation, so the pre-computation is performed only once. More generally, the projection can be updated whenever the spatial pattern changes (e.g., between successive pushes/insonications at different foci), with only $O(N)$ scaling required during the intervening time steps.

Each pre-computation requires one CG solve at cost $O(N \cdot n_{\text{iter}})$, while the per-step scaling costs only $O(N)$. When the spatial pattern is fixed, the total projection cost is $O(N \cdot n_{\text{iter}} + N \cdot N_t)$ compared with $O(N \cdot n_{\text{iter}} \cdot N_t)$ for the general per-step solver. For practical problem sizes—for example, $N = 106^3 \approx 1.2 \times 10^6$ points and $N_t = 85$ steps—the pre-computed projection reduces total simulation time from $O(10^2\text{--}10^3)$ s to $O(10)$ s, an improvement of one to two orders of magnitude. Both the general per-step solver and the pre-computed variant are implemented; the choice is made automatically based on whether the user supplies a separable source.

3.4 Sequential acoustic–elastic coupling

Because the acoustic field evolves on timescales three orders of magnitude faster than shear propagation, the two problems can be solved sequentially: the acoustic pressure and particle velocity fields are computed first, a body force distribution is derived from them, and this force is supplied as a prescribed source term to Eq. (16). This sequential coupling is the standard workflow in SWE simulation [16, 22].

The acoustic field can be obtained from measurement, analytical calculation, or with any suitable numerical solver such as Field II, Fullwave [44] or the angular spectrum approach [45].

Two formulations are used to compute the acoustic radiation body force from the acoustic field. The plane-wave approximation gives a scalar force along the beam axis [10, 11]:

$$b_0 = \frac{2\alpha_{\text{Np}}}{c} I_{\text{SPPA}}, \quad I_{\text{SPPA}} = \frac{1}{\rho c} \frac{\sum_{i=1}^{N_{\text{total}}} p_i^2}{N_{\text{pulse}}}, \quad (38)$$

where α_{Np} is the absorption in Np/m at the transmit frequency. The plan wave approximation is easier to implement for non-vectorial estimates of the pressure wave-field. However, for converging beams the plane-wave assumption breaks down near the focal zone and a more accurate approach uses the time-averaged Poynting vector $\mathbf{I} = \langle p \mathbf{v} \rangle$:

$$f_i = \frac{2\alpha_{\text{Np}}}{c} I_i, \quad I_i = \langle p v_i \rangle, \quad i \in \{x, y, z\}, \quad (39)$$

yielding three spatially varying force components.

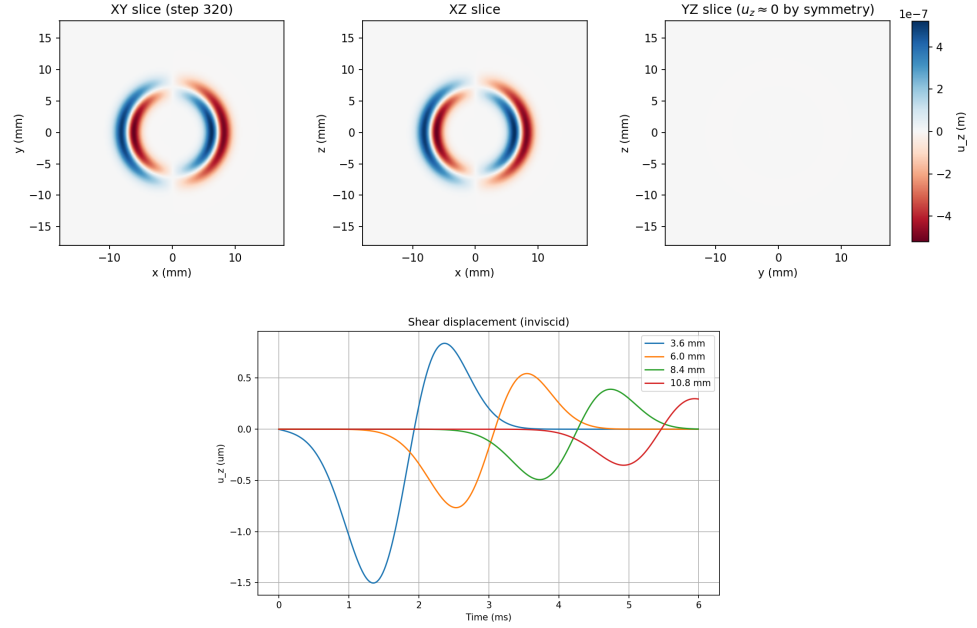


Figure 1. Non-viscous wavespeed validation. **Top:** Shear-wave field snapshot (u_z): x - y slice (left), x - z slice (middle), y - z slice at $x = 0$ (right). **Bottom:** Displacement traces (u_z) at four sensor offsets (3.6, 6.0, 8.4, 10.8 mm). Wavespeed is estimated from peak arrival times.

3.5 Boundary and initial conditions

In the current implementation, the computational domain is terminated with a fixed (Dirichlet) boundary condition by setting displacement to zero on the outer layer. This choice simplifies the staggered-grid update and provides a well-posed finite domain problem, but it can introduce reflections that contaminate late-time arrivals. Absorbing boundary treatments such as perfectly matched layers (PML) [46, 47] can reduce such artifacts when late-time fidelity is required. Accordingly, validation and analysis are restricted to early and intermediate propagation times and/or to sensor offsets that are not dominated by boundary interactions. The initial state is commonly specified as a divergence-free velocity field constructed from a vector potential (a curl-based initialization), which preferentially excites shear motion while suppressing compressional content.

4 Results

4.1 Non-viscous wavespeed recovery

To isolate elastic wavespeed recovery, the first validation disables viscosity ($\eta = 0$) and uses a curl-based vector-potential source to excite a divergence-free initial velocity field. This source is constructed from a Gaussian envelope in the x - y plane, applied through the curl of a vector potential so that the resulting velocity field is solenoidal by construction. The curl initialization preferentially excites shear motion while suppressing compressional content, providing a test of the shear propagation numerics without confounds that may arise from longitudinal waves or from the Helmholtz projection step.

The simulation domain was implemented as a cubic 180^3 grid ($L_x = L_y = L_z = 36$ mm, $\Delta x = 0.2$ mm) with homogeneous soft-tissue parameters ($\rho = 1000$ kg/m³, $c_s = 2.0$ m/s). Figure 1 shows the resulting shear-wave displacement field (u_z) at a representative time step in three orthogonal slice planes, along with u_z traces recorded at four sensors placed at offsets of 3.6, 6.0, 8.4, and 10.8 mm from the source along the depth axis. The wave propagates outward in the x - y and x - z planes as expected. The y - z slice at $x = 0$ is near zero because the curl-based initialization drives motion in the x - z plane, and the center x -plane is a nodal surface for u_z by symmetry. Wavespeed was estimated from peak arrival times in the sensor traces.

The target shear speed is 2.000 m/s (from $\mu = \rho c_s^2$). The mean recovered speed is 1.976 m/s ($\sim 1.2\%$ below target) and the distance-time linear-fit speed is 1.998 m/s ($< 0.1\%$ error), confirming that the shear-wave propagation physics and numerics are correct in this non-viscous benchmark regime.

Table 1. Grid convergence of the full-waveform trace L_2 self-convergence error (relative L_2 norm of the velocity traces against the finest grid, $n = 180$; fixed 36 mm domain, fixed 3 mm physical source).

n	Δx (μm)	trace L_2 error
60	600	2.65%
90	400	1.08%
120	300	0.47%
150	240	0.17%
180	200	— (reference)

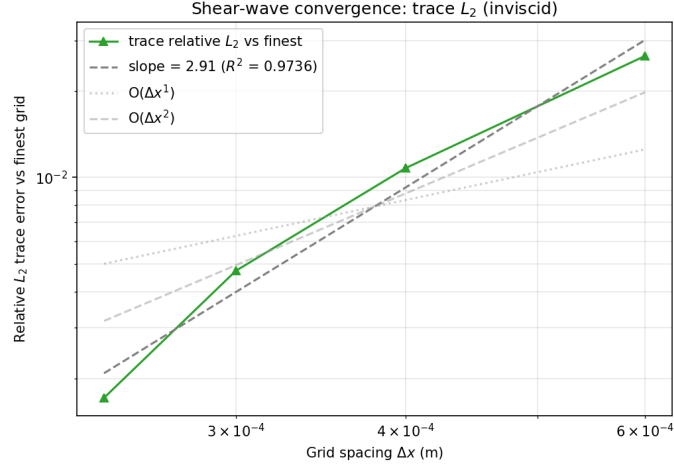


Figure 2. Grid convergence of the full-waveform trace L_2 self-convergence error (relative L_2 error of velocity traces vs the finest grid) against grid spacing (log-log), for a fixed 3 mm physical source. The fitted global slope is $p \approx 2.9$ ($R^2 = 0.97$); dashed reference lines show $O(\Delta x)$ and $O(\Delta x^2)$.

4.2 Grid convergence study

An inviscid convergence sweep was performed with fixed physical domain size ($L_x = L_y = L_z = 36$ mm) and fixed physical sensor distances, over a five-level refinement range $n \in \{60, 90, 120, 150, 180\}$ chosen so that the sensor distances fall exactly on grid nodes at every level. So that each grid discretizes the *same* continuum problem, the Gaussian excitation is held at a fixed *physical* width ($\sigma_{xy} = 3$ mm, i.e. 5 grid points at the coarsest level and proportionally more as the grid refines) and the velocity-damping coefficient is scaled per time step so that the cumulative damping over the fixed time window is grid-independent. Holding the source width at a fixed number of grid points instead—so that its physical size shrinks under refinement—conflates discretization error with a changing source, and is avoided here.

Convergence is quantified by the full-waveform *trace* L_2 error: the relative L_2 norm of the velocity-trace difference at each sensor (interpolated onto a common time base), averaged across sensors, measured against the finest grid ($n = 180$) as a self-convergence reference. Unlike a scalar arrival-time speed, it captures the whole waveform. Table 1 lists the error at each level and Fig. 2 shows it against grid spacing. The error falls monotonically from 2.7% to 0.2%. Fitting trace- $L_2 \propto \Delta x^p$ in log-log space gives a global slope $p = 2.9$ ($R^2 = 0.97$); the well-separated levels give per-interval orders $p \approx 2.2$ – 2.9 , with the pair nearest the reference inflated by the self-convergence reference (a known bias of self-convergence studies). This confirms convergence at least as fast as the formal second-order accuracy of the 7-point Laplacian stencil and leapfrog time integration.

We deliberately do *not* use the arrival-time wavespeed error as a grid-convergence metric. With the source width held physically fixed, the peak-picked wavespeed reflects a source-width- and near-field-dependent estimation bias that does not vanish under refinement, so it is not a clean measure of discretization error. The accuracy of the recovered wavespeed itself is instead established by the non-viscous benchmark (Fig. 1, linear-fit speed within $<0.1\%$).

4.3 Kelvin–Voigt attenuation benchmark

To avoid known bias from finite 3D pulse-domain effects, viscosity validation uses a dedicated 1D harmonic benchmark, which better matches Kelvin–Voigt analytic assumptions (single frequency, effectively plane-wave propagation, homogeneous medium). To align with Kelvin–Voigt theory and cover a soft tissue range, the benchmark was run on $\eta \in [0, 1.5]$ Pa·s with $\Delta\eta = 0.1$ Pa·s. The high- η configuration uses (i) adaptive sensor placement based on the expected e-folding length, (ii) adaptive

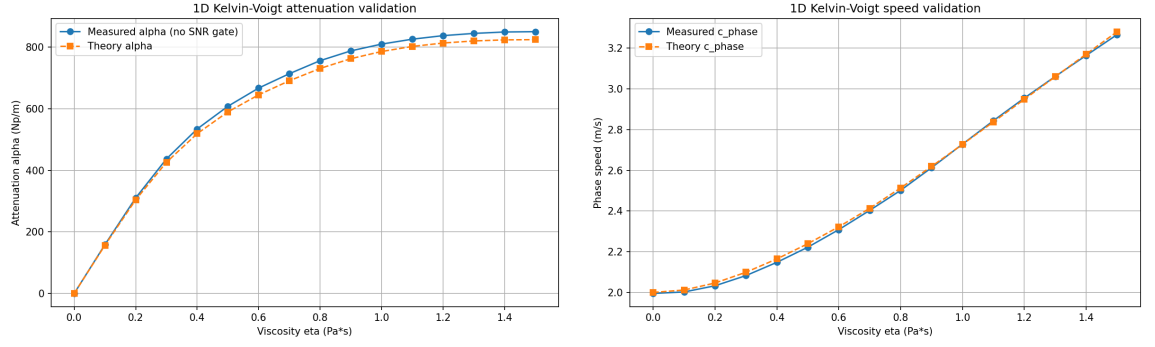


Figure 3. 1D harmonic Kelvin–Voigt benchmark over $\eta \in [0, 1.5]$ Pa·s with adaptive sensor placement and frequency scheduling. **Left:** Attenuation (measured vs theory). **Right:** Phase speed (measured vs theory).

Table 2. 1D harmonic Kelvin–Voigt benchmark: measured vs. theoretical attenuation and phase speed at selected viscosities. Errors are relative to the exact Kelvin–Voigt dispersion relation at the scheduled drive frequency.

η (Pa·s)	α_{meas} (Np/m)	α_{th} (Np/m)	α error (%)	c_{meas} (m/s)	c_{th} (m/s)	c error (%)
0.10	158.5	156.4	+1.3	2.002	2.012	−0.5
0.30	436.8	425.8	+2.6	2.083	2.099	−0.8
0.50	607.8	589.5	+3.1	2.222	2.239	−0.8
1.00	809.7	785.6	+3.1	2.729	2.728	< 0.1
1.50	850.0	824.2	+3.1	3.267	3.281	−0.4

drive amplitude scaling, and (iii) a smooth frequency schedule $f_0(\eta)$ guided by dimensional analysis.

Define the loss parameter and dimensionless attenuation

$$\text{De}_v = \frac{\omega\eta}{\mu}, \quad \Pi_\alpha = \frac{\alpha c_s}{\omega}, \quad c_s = \sqrt{\mu/\rho}. \quad (40)$$

For weak-to-moderate loss ($\text{De}_v \ll 1$),

$$\Pi_\alpha \approx \frac{\text{De}_v}{2} \Rightarrow \alpha \approx \frac{\eta\omega^2}{2\rho c_s^3}. \quad (41)$$

A key challenge for attenuation measurement at high viscosity is that the e-folding length $\ell_e = 1/\alpha$ shrinks rapidly. At $\eta = 1.5$ Pa·s the theoretical $\alpha \approx 824$ Np/m gives $\ell_e \approx 1.2$ mm. Sensors placed beyond a few e-folding lengths measure only noise, biasing the attenuation fit low. Sensor positions are thus adapted per η based on the theoretical attenuation at the scheduled drive frequency.

Table 2 presents representative results from the 16-point viscosity sweep. Across the full range $\eta \in [0.1, 1.5]$ Pa·s, the measured attenuation tracks the Kelvin–Voigt prediction with a mean absolute error of $\sim 3\%$, while the phase speed agrees to within $< 0.5\%$. Both measured quantities increase monotonically with viscosity, consistent with the analytical dispersion relation. The results demonstrate robust attenuation recovery across the full viscosity range, confirming the implicit viscous time-stepping scheme.

4.4 QIBA elastic phantom benchmark

The preceding validations use idealized source configurations (curl-based impulse, harmonic plane wave) to isolate individual solver properties. This section validates the complete end-to-end pipeline (acoustic simulation, radiation force computation, and shear wave propagation) against the standardized digital phantoms defined by the RSNA QIBA Ultrasound Shear Wave Speed Biomarker Committee [16]. These phantoms were designed for inter-solver comparison and have been validated with both LS-DYNA and Abaqus finite element codes.

Four homogeneous elastic phantoms are simulated with shear moduli $G \in \{1, 2, 5, 10\}$ kPa, density $\rho = 1000$ kg/m³, and Poisson’s ratio $\nu = 0.495$ (nearly incompressible), matching the QIBA specification. The acoustic radiation force is modeled as a three-dimensional Gaussian body force following the convention used in the QIBA FEM pipeline [16]:

$$b(\mathbf{x}) = A \exp\left[-\left(\frac{x - x_f}{\sigma_x}\right)^2 - \left(\frac{y}{\sigma_y}\right)^2 - \left(\frac{z}{\sigma_z}\right)^2\right], \quad (42)$$

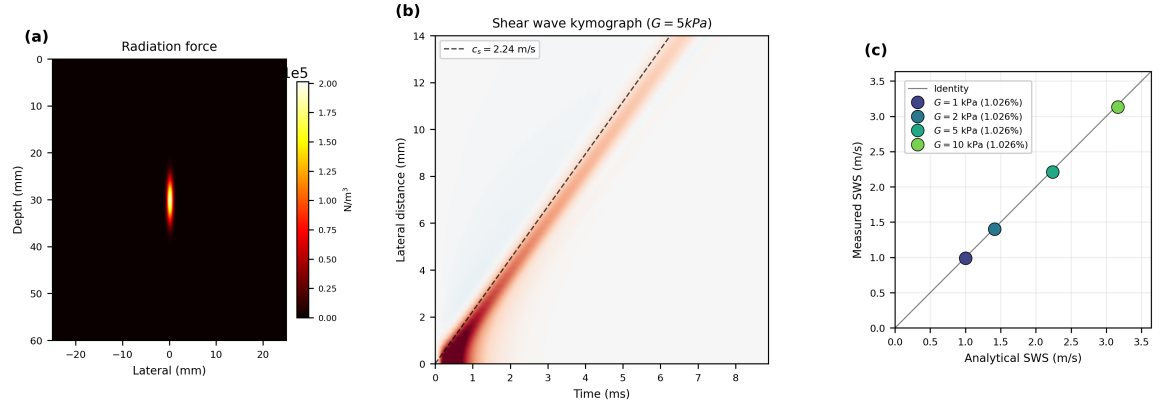


Figure 4. QIBA elastic phantom validation using a Gaussian body force excitation matching the QIBA convention. (a) Gaussian radiation force distribution (Eq. 42), with the focal spot centered at 30 mm depth. (b) Space–time kymograph of axial displacement for the $G = 5$ kPa phantom, where the dashed line indicates the analytical shear wave speed $c_s = 2.24$ m/s. (c) Recovered shear wave speed vs. analytical value for all four QIBA elastic phantoms ($G = 1$ –10 kPa), all configurations recover the wavespeed to within $\sim 1\%$.

where $x_f = 30$ mm is the focal depth and the $1/e$ widths are derived from the beam physics of the QIBA curvilinear array ($f/2.0$, 3 MHz): $\sigma_y \approx 0.62$ mm (lateral), $\sigma_x \approx 4.3$ mm (axial), and $\sigma_z \approx 1.1$ mm (elevational, from the 50 mm lens focus). The push duration is 167 μ s (500 cycles at 3 MHz). The computational grid uses $\Delta x = 0.25$ mm with a domain of $60 \times 50 \times 50$ mm³ ($240 \times 200 \times 200$ nodes). For each phantom, the solver is run with $\eta = 0$ (pure elastic) and the shear wave group velocity is estimated by cross-correlation of velocity traces at lateral offsets of 4–14 mm from the push axis.

Figure 4(a) shows the computed radiation force distribution, exhibiting the characteristic tight focal spot of an $f/2.0$ focused beam. The space–time kymograph in Figure 4(b) displays the displacement field for the $G = 5$ kPa phantom as a function of lateral distance and time. The shear wave front propagates at a slope consistent with the analytical speed $c_s = \sqrt{G/\rho} = 2.24$ m/s (dashed line). Figure 4(c) summarizes the recovered shear wave speed for all four phantoms. Errors relative to the analytical value are $\sim 1.0\%$ for all four configurations ($G = 1$ –10 kPa), confirming accurate wavespeed recovery across a tenfold range of shear moduli. This level of agreement is well within the inter-solver variability reported in the QIBA round-robin comparison [16].

5 Transcranial Application Demonstrations

Having validated the shear wave solver on homogeneous benchmark media, this section demonstrates a complete acousto–elastic pipeline (Section 3.4) applied to transcranial ultrasound through a realistic human skull. Two transducer configurations are considered: a 1024-element sparse imaging array emitting an ARF push [48] and the focused ultrasound Philips TIPS curved-bowl transducer emitting a neuromodulatory sequence. Both simulations use the same *ex vivo* micro-CT skull specimen (Halle temporal bone) [49] and share the same four-stage workflow: (i) acoustic transmit through the skull (GPU FDTD), (ii) computation of the acoustic radiation force, (iii) shear wave FDTD simulation of viscoelastic brain tissue displacement, and (iv) visualization of all intermediate and final fields.

5.1 Sparse phased array through micro-CT skull

The first demonstration uses a 1024-element sparse phased array with 750 μ m element pitch [48]. Element positions are loaded from an HDF5 connector file and mapped onto the simulation grid (66×66 mm lateral–elevational aperture). Per-element focusing delays steer the beam to a 50 mm depth focal point. The heterogeneous skull medium is constructed from a micro-CT NRRD volume of the Halle temporal bone specimen [49]. Hounsfield units are mapped to spatially varying sound speed (1540–2900 m/s), density (1000–2200 kg/m³), and attenuation coefficient following established CT-to-acoustic mappings [50, 7]. The background tissue is assigned standard soft-tissue properties ($c = 1540$ m/s, $\rho = 1000$ kg/m³, $\alpha_0 = 0.5$ dB cm^{−1} MHz^{−1}) [19]. The acoustic simulation uses the Fullwave GPU FDTD solver [51] with exponential attenuation at $f_0 = 1$ MHz, source pressure amplitude of 1.5 MPa, and a sparse-grid sensor ($4\times$ downsampling in each spatial dimension). Figure 5 shows the acoustical medium property maps and three pressure propagation snapshots as the focused beam propagates through the skull and converges at the 50 mm focal zone. The velocity-based Poynt-

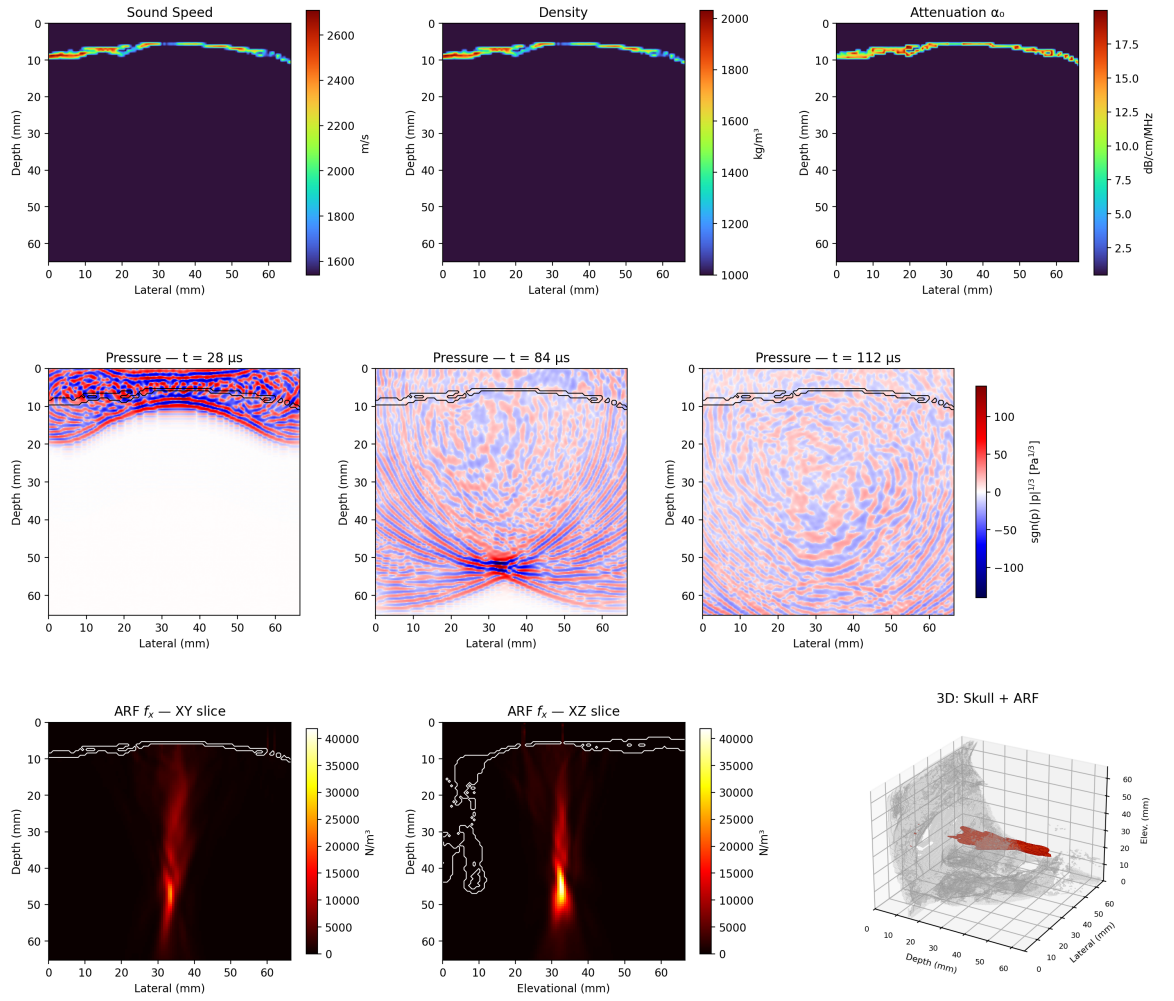


Figure 5. Acoustical medium, pressure propagation, and acoustic radiation force for the sparse array transcranial configuration. **Top row:** Sound speed, density, and attenuation coefficient α_0 derived from micro-CT skull data (x - y slices at full resolution). **Middle row:** Pressure propagation snapshots showing the focused beam emerging from the transducer, propagating through the skull (black contour), and converging at the focal zone. **Bottom row:** Depth-component ARF (f_x) in x - y and x - z slices (brain-masked, skull contour in white), with a 3D rendering of the ARF isosurface (red) within the skull geometry (gray).

ing vector formulation Eq. (39) is used with pulse normalization corrections to compute the acoustic radiation force. Non-brain regions (coupling medium, skull interior, and pre-skull tissue) are masked to zero so that only the intracranial force field drives the shear wave simulation. The depth component gives $f_x^{\max} \approx 41,800 \text{ N/m}^3$, with lateral and elevational components $|f_y|^{\max} \approx 7,000 \text{ N/m}^3$ and $|f_z|^{\max} \approx 6,400 \text{ N/m}^3$. Figure 5 (bottom row) shows the depth component of the resulting body force distribution. The pre-skull coupling medium and skull interior are masked so that only the brain-side force field is displayed. A 3D rendering of the ARF isosurface (red) within the skull geometry (gray) is included for spatial context.

The shear wave simulation uses the staggered-grid FDTD solver with brain tissue parameters: shear wave speed $c_s = 2.14 \text{ m/s}$ at 75 Hz, density $\rho = 1000 \text{ kg/m}^3$, giving shear modulus $\mu = \rho c_s^2 \approx 4580 \text{ Pa}$. Kelvin-Voigt viscosity is set to $\eta = 0.5 \text{ Pa}\cdot\text{s}$. All three ARF components are applied as forcing terms via the separable temporal envelope, enabling a single pre-computed Helmholtz projection. Peak displacement reaches $|u_x|^{\max} \approx 1.7 \mu\text{m}$. Figure 6 shows the displacement magnitude field expanding radially from the focal spot over 0.7–3.0 ms, and displacement time traces over 10 ms at the focal spot and at lateral and depth offsets of 3.1, 6.2, and 9.2 mm, confirming the expected shear wave propagation speed and amplitude decay with distance.

5.2 TIPS curved-bowl transducer through micro-CT skull

The second demonstration uses the Philips TIPS (Therapeutic Imaging and Phased array Sonication) transducer, a curved-bowl annular phased array designed for focused ultrasound therapy and

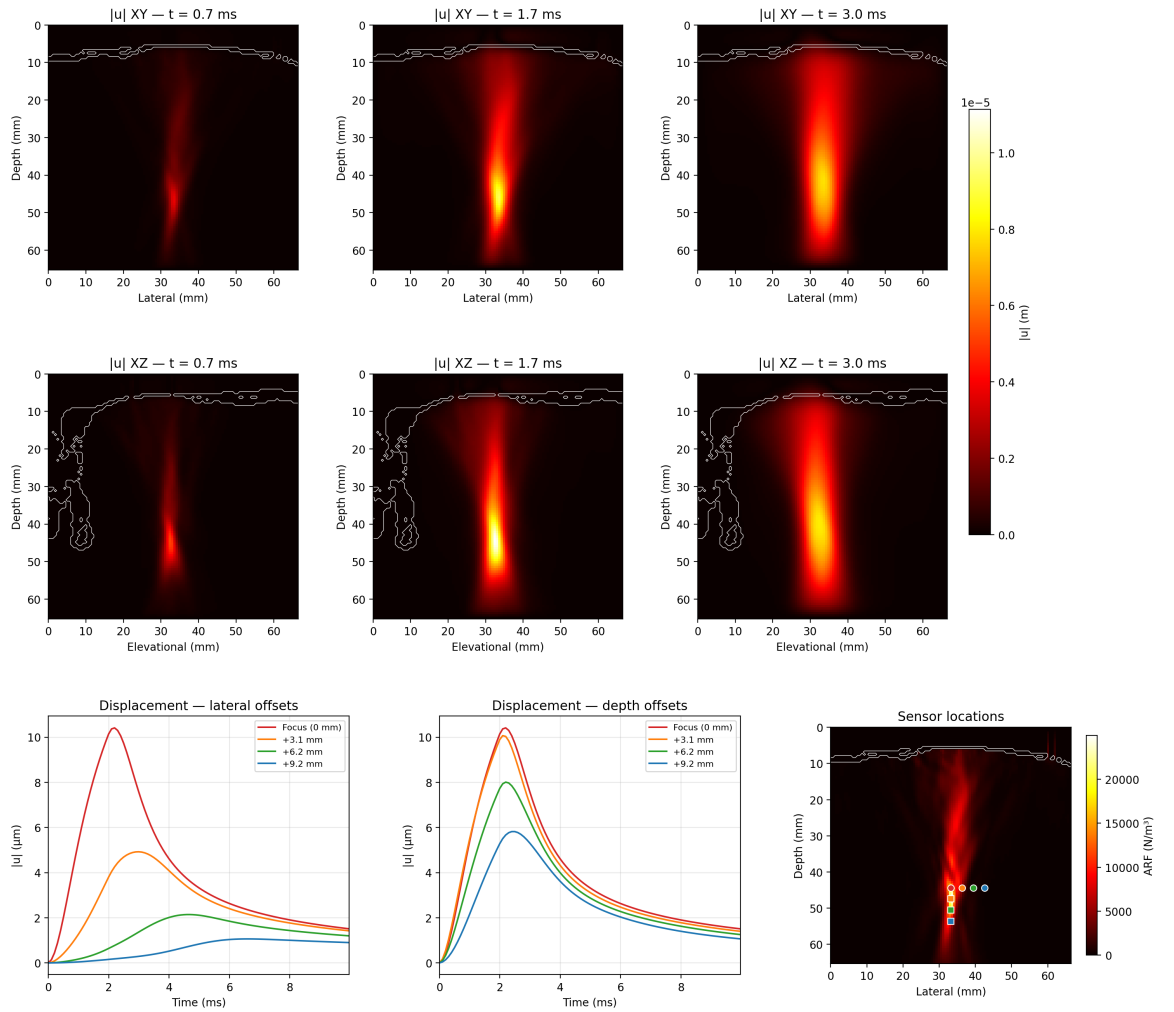


Figure 6. Shear wave displacement for the sparse array configuration. **Top row:** displacement magnitude $|u|$ in $x-y$ slices at center elevation. **Middle row:** $|u|$ in $x-z$ slices at center lateral position. White contours indicate the skull boundary. **Bottom row:** displacement time traces at the focal spot and at lateral offsets (left) and depth offsets (centre) from the ARF peak. Sensor locations are shown on the ARF map (right, circles = lateral, squares = depth offsets).

neuromodulation. The TIPS transducer is modeled as a spherical-cap bowl: radius of curvature $R = 80$ mm, inner aperture radius $r_{\text{in}} = 20.5$ mm, outer aperture radius $r_{\text{out}} = 46$ mm (92 mm full diameter), 8 concentric annular elements of equal radial width, and a bowl thickness of 3 voxel layers along the surface normal.

The heterogeneous skull medium uses the same Halle temporal bone micro-CT specimen as Section 5.1. Because the TIPS bowl surface extends further in depth, the skull slab is shifted deeper to avoid geometric intersection. The acoustic simulation uses 400 kPa source pressure with 3-cycle Gaussian-modulated sinusoidal pulses at $f_0 = 1$ MHz. Figure 7 summarises the full TIPS transcranial simulation pipeline. Panels (a,b) show the transducer placement on axial and sagittal micro-CT slices of the Halle skull, with the annular TIPS aperture (gold) and geometric focal spot (cyan star) overlaid. Panel (c) renders the skull slab (semi-transparent), TIPS bowl (gold), and ARF isosurface (hot colourmap) in 3D. The geometrically focused bowl produces a converging wavefront that propagates through the skull and focuses at 50 mm depth. Figure 7(d,e) shows the pressure field at $t = 28 \mu\text{s}$ (converging wavefront) and $t = 104 \mu\text{s}$ (post-focal reflections). The acoustic radiation force is computed using the pressure-based plane-wave approximation Eq. (38) with pulse normalization corrections. The peak body force after brain masking is $b_0^{\text{max}} \approx 22,600 \text{ N/m}^3$. Figure 7(f) shows the depth component of the radiation force in the x - y plane with brain masking and skull contour overlay. The shear wave simulation uses the same brain tissue parameters as Section 5.1. The pulsed forcing protocol uses a 50% duty cycle (1 ms on, 1 ms off) sustained over 50 ms (25 pulses). Peak displacement reaches $|u_x|^{\text{max}} \approx 5 \mu\text{m}$ at the focal spot. Figure 7(h) shows displacement time traces at the focal spot and at lateral offsets of 3.1, 6.2, and 9.2 mm. The volumetric acoustic strain $\varepsilon = -p/K$ (where $K = \rho c^2$ is the bulk modulus) and its spatial gradient are computed directly from the pressure time series without the pulse-length normalization used for radiation force. Peak acoustic strain reaches 8.8×10^{-4} and peak strain gradient magnitude reaches 1.5 m^{-1} . Figure 7(g) shows the peak strain field. These quantities are relevant to flexoelectric coupling mechanisms proposed for ultrasonic neuromodulation [52].

6 Discussion

6.1 Physical assumptions

The solver is derived under a small-strain linearization i.e. displacement gradients are assumed to be small, and the stress-strain relationship is linear. This is typically appropriate for ultrasound radiation-force excitations in SWE [1, 2], where observed tissue motion is commonly in the micrometer to tens-of-micrometers regime.

Material behavior is modeled as isotropic with an effectively homogeneous (or piecewise homogeneous) shear response, represented by a scalar shear modulus $\mu(\mathbf{x})$ and density $\rho(\mathbf{x})$ [34, 19]. This approximation is often reasonable in parenchymal soft tissues at the spatial resolution used for SWE processing [3], but it can break down in anisotropic or strongly heterogeneous structures such as skeletal muscle, tendon, white-matter tracts, and near rigid interfaces (e.g., skull).

A central modeling choice is to represent shear-dominant propagation while omitting explicit compressional (longitudinal) elastodynamics. This is motivated by the near-incompressibility of soft tissue ($K \gg \mu$). The tradeoff is that this reduced model does not capture near-source pressure-wave content, P-S mode conversion at interfaces, nonlinear shear or shear shock dynamics [53, 54, 55], or any diagnostic relying on volumetric strain.

Viscous attenuation is represented using a Kelvin-Voigt shear viscosity [38, 39]. This single-parameter form is useful for capturing qualitative attenuation and pulse broadening, but soft tissue often exhibits power-law or fractional viscoelastic behavior [56]. A constant η may be insufficient for broadband inversions or dispersion-sensitive analyses.

Finally, the present implementation enforces fixed (zero-displacement) boundaries. Practical mitigation strategies include using larger computational domains, restricting analysis to earlier times, or implementing absorbing layers (e.g., PML) [46, 47].

6.2 Computational cost

The reduced FDTD formulation is computationally lightweight compared to general-purpose FEM solvers. For the TIPS transcranial demonstration (Section 5.2), the shear wave simulation uses a $129 \times 181 \times 181$ grid (4.2×10^6 points) and 3009 time steps. With pre-computed Helmholtz projection, the shear FDTD completes in approximately 4 minutes on a single CPU core (NumPy, no GPU acceleration). The QIBA benchmark ($240 \times 200 \times 200$ grid, 9.6×10^6 points) runs in under 10 minutes under the same conditions. By comparison, FEM approaches to ARF-driven shear wave simulation using commercial codes such as LS-DYNA typically require hours of computation on comparable 3D grids [23, 16]. Green's function methods [22] can be faster for homogeneous media

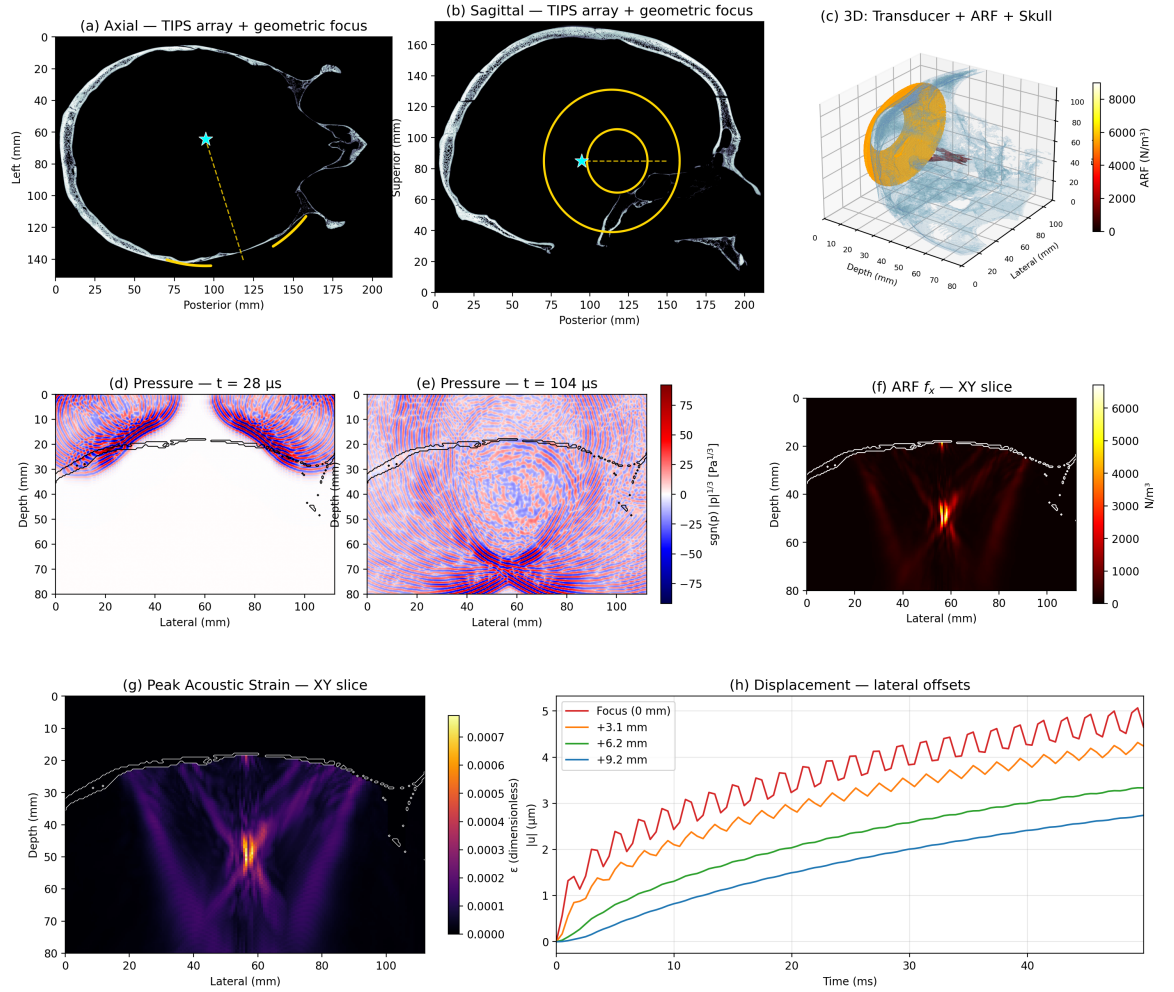


Figure 7. Composite TIPS transcranial shear wave simulation. **(a,b)** Axial and sagittal micro-CT slices showing the TIPS annular aperture (gold rings) and geometric focal spot (cyan star). **(c)** 3D rendering of the skull slab (semi-transparent blue), TIPS bowl transducer (gold), and ARF isosurface (coloured by intensity). **(d,e)** Pressure propagation at $t = 28$ and $104 \mu\text{s}$ (skull contour in black). **(f)** Brain-masked ARF depth component b_0 in the x - y plane (skull contour in white). **(g)** Peak volumetric acoustic strain ϵ in the x - y plane. **(h)** Displacement magnitude time traces at the focal spot and at lateral offsets from the ARF peak.

Table 3. Frequency dependence of ARF body force at fixed focal intensity ($I_{\text{SPPA}} = 1000 \text{ W/cm}^2$, $\alpha_0 = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$, $c = 1540 \text{ m/s}$).

f_0 (MHz)	α_{NP} (Np/m)	b (N/m ³)
1	5.8	750
3	17.3	2 250
5	28.8	37 400
6.7	38.6	50 100

and sparse observation points, but do not naturally extend to full-field output or strong spatial heterogeneity. The present solver occupies a middle ground: it provides full-field output with spatial heterogeneity at a fraction of the cost of FEM, while remaining a general-purpose grid solver rather than a point-wise evaluator.

6.3 Frequency dependence of ARF magnitude

Because absorption in soft tissue scales approximately linearly with frequency, the ARF magnitude depends strongly on f_0 even at the same focal intensity (Table 3). In clinical soft-tissue ARFI at 5–7 MHz, focal intensities of 500–2000 W/cm² produce body forces on the order of 10^4 – 10^5 N/m^3 , yielding tissue displacements of 1–20 μm . For transcranial applications at lower frequencies (0.5–1 MHz), the absorption per unit path length is roughly an order of magnitude lower, and the skull attenuates the beam to roughly 30–50% of its free-field value. The net effect is that body forces at 1 MHz are typically 10^2 – 10^4 N/m^3 , and the resulting brain tissue displacements are sub-micrometer to a few micrometers.

6.4 Applicability and limitations

The reduced model is most appropriate in regimes with small displacements and approximately linear response [1], in nearly incompressible soft tissue where SWE biomarkers depend primarily on the shear-wave speed [20, 21, 2], in moderate propagation ranges where boundary reflections are not dominant, and in analyses that are effectively narrowband such that a Kelvin–Voigt damping term provides an adequate approximation [38]. The assumptions are more likely to fail in strongly anisotropic or heterogeneous media, in problems where compressional waves must be represented explicitly, in regimes where nonlinear shear or shear shock formation is significant [53, 54], in broadband rheological inversions, in domains where fixed-wall reflections strongly influence arrival times, and in settings where poroelasticity is needed.

7 Conclusion

This manuscript derived a reduced shear elastodynamic formulation and an explicit FDTD implementation for ultrasound-driven shear wave propagation in soft tissue. The governing equation is obtained by systematically applying near-incompressibility, small-strain linearization, Helmholtz decomposition, and solenoidal force projection to the full Navier equations, yielding a Kelvin–Voigt shear wave equation that isolates the transverse dynamics relevant to radiation-force-induced motion.

Validation in homogeneous benchmark media demonstrates accurate recovery of the imposed shear wavespeed under controlled conditions ($<0.1\%$ error at the finest grid), second-order convergence under grid refinement, and agreement with analytical Kelvin–Voigt attenuation and phase speed predictions ($\sim 3\%$ and $<0.5\%$, respectively). An end-to-end benchmark against the RSNA QIBA elastic phantom standard confirms accurate shear wave speed recovery ($\sim 1\%$ error) across a tenfold range of shear moduli, validating the complete acoustic–elastic pipeline including radiation force computation. The pre-computed Helmholtz projection for separable body forces reduces simulation time by one to two orders of magnitude in typical ARF-driven scenarios.

The principal limitations are those implied by the model reduction which include compressional dynamics, mode conversion at interfaces, anisotropy, strong heterogeneity, broadband frequency-dependent viscoelasticity, and late-time reflection artifacts are not fully represented. Within its intended scope, the reduced solver provides a lightweight, purpose-built tool for multiphysics ultrasound workflows where shear wave generation, propagation, and measurement are the dominant quantities of interest [1, 17, 2, 8, 21]. End-to-end transcranial demonstrations with sparse phased array and curved-bowl transducer geometries confirm integration with GPU acoustic simulators and radiation force computation through realistic skull models, producing shear displacements of 1–3 μm consistent with clinical ARFI measurements. The computational efficiency of the reduced solver

broadens applications to inverse problems, iterative methods, and machine learning data generation tasks.

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Author contributions

Gianmarco F Pinton: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

Data availability

The simulation code and validation scripts are available at <https://github.com/pinton-lab/shearwave>.

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